



Forces, Friction and Inclined Planes

· Mark Scheme

Mechanics Chapter 6: Forces, Friction and Inclined Planes. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Resolving forces

- 1 $10 \cos 30^\circ = 8.66 \text{ N}$ B1; $10 \sin 30^\circ = 5 \text{ N}$ B1.
- 2 Components: $6 + 8 \cos 60^\circ = 10$, $8 \sin 60^\circ = 6.93$ M1; magnitude 12.2 N A1; angle 34.7° A1.
- 3 $P \cos \theta = 5$, $P \sin \theta = 12$ M1; $P = 13$ A1; $\theta = 67.4^\circ$ A1.
- 4 $R(\rightarrow)$: $T_1 \cos 30^\circ = T_2 \cos 60^\circ$ M1; $R(\uparrow)$: $T_1 \sin 30^\circ + T_2 \sin 60^\circ = 20$ M1; $T_1 = 10 \text{ N}$ A1; $T_2 = 17.3 \text{ N}$ A1.
- 5(a) Weight 19.6 N down, T along the string, P horizontal B1.
- 5(b) $R(\uparrow)$: $T \cos 25^\circ = 19.6$ M1 A1; $T = 21.6 \text{ N}$ A1.
- 5(c) $P = T \sin 25^\circ$ M1; $= 9.14$ A1.
- 5(d) The angle with the vertical increases B1.

Section B – Friction

- 6 $F_{\max} = 0.3 \times 49$ M1; $= 14.7 \text{ N}$ A1.
- 7 $F_{\max} = 0.4 \times 98 = 39.2 \text{ N}$ M1; $30 < 39.2$, so the box does not move A1; friction $= 30 \text{ N}$ A1.
- 8 $F = 0.25 \times 39.2 = 9.8 \text{ N}$ B1; $30 - 9.8 = 4a$ M1; $a = 5.05 \text{ ms}^{-2}$ A1.
- 9 $R = 78.4 - 40 \sin 30^\circ = 58.4$ M1 A1; $F = 40 \cos 30^\circ = 34.6$ B1; $\mu = \frac{34.6}{58.4} = 0.593$ A1.
- 10(a) $R = 58.8 + P \sin 20^\circ$ M1; $P \cos 20^\circ = 0.35R$ M1; $P(\cos 20^\circ - 0.35 \sin 20^\circ) = 20.58$ M1; $P = 25.1$ A1.
- 10(b) $R = 58.8 + 40 \sin 20^\circ = 72.5$ B1; $F = 0.35R = 25.4$ B1; $40 \cos 20^\circ - 25.4 = 6a$ M1; $a = 2.04 \text{ ms}^{-2}$ A1.

Section C – Inclined planes

- 11 $a = g \sin 30^\circ$ M1; $= 4.9 \text{ ms}^{-2}$ A1.
- 12 $P = 5g \sin 20^\circ = 16.8 \text{ N}$ M1 A1; $R = 5g \cos 20^\circ = 46.0 \text{ N}$ A1.
- 13 $R = 3g \cos 25^\circ$ B1; $3g \sin 25^\circ - 0.3R = 3a$ M1; $a = 1.48 \text{ ms}^{-2}$ A1.
- 14 (a) $R = mg \cos \alpha$ B1; $F = mg \sin \alpha$ and $F = \mu R$ M1; $\mu = \tan \alpha$ A1. (b) $\tan 35^\circ = 0.700$ B1.
- 15(a) Friction and the weight component both act down the plane M1; $4a = 4g \sin 30^\circ + 0.2 \times 4g \cos 30^\circ$ A1; deceleration 6.60 ms^{-2} A1.
- 15(b) $0 = 64 - 2(6.597)s$ M1; $s = 4.85 \text{ m}$ A1.
- 15(c) Compare $mg \sin 30^\circ = 19.6 \text{ N}$ with $F_{\max} = 0.2 \times 4g \cos 30^\circ = 6.79 \text{ N}$ M1; $19.6 > 6.79$, so it slides back down A1.
- 15(d) $4a = 4g \sin 30^\circ - 0.2 \times 4g \cos 30^\circ$ M1; $a = 3.20 \text{ ms}^{-2}$ A1.

Section D – Connected particles with friction

- 16 $F = 0.2 \times 3g$ M1; $= 5.88 \text{ N}$ A1.
- 17 $2g - 5.88 = 5a$ M1; $a = 2.74 \text{ ms}^{-2}$ A1; $T = 2(g - 2.744) = 14.1 \text{ N}$ A1.
- 18 Friction: $0.25 \times 10g = 24.5 \text{ N}$ in total B1; $40 - 24.5 = 10a$, $a = 1.55 \text{ ms}^{-2}$ M1; A: $T - 9.8 = 4 \times 1.55$, $T = 16.0 \text{ N}$ A1.
- 19 Plane: $T - 3g \sin 30^\circ - 0.2 \times 3g \cos 30^\circ = 3a$ M1; hanging: $4g - T = 4a$ M1; $a = 2.77 \text{ ms}^{-2}$ A1; $T = 28.1 \text{ N}$ A1.



20(a) $\sin \alpha = 0.6$, $\cos \alpha = 0.8$ *B1*; $R = 5g(0.8) = 4g$, $F = 0.5R = 2g$ down the plane *M1*; $mg = 5g(0.6) + 2g$ *M1*; $m = 5$ *A1*.

20(b) Friction now acts up the plane: $mg = 3g - 2g$ *M1*; $m = 1$ *A1*.

20(c) $8g - 3g - 2g = 13a$ *M1 A1*; $a = 2.26 \text{ ms}^{-2}$ *A1*.

Roar-level challenge

21 $R = 98 - P \sin \theta$, $P \cos \theta = 0.75R$ *M1*; $P(\cos \theta + 0.75 \sin \theta) = 73.5$ *A1*; $\cos \theta + 0.75 \sin \theta = 1.25 \cos(\theta - 36.9^\circ)$ *M1*; least P when $\theta = 36.9^\circ$ *A1*; $P = \frac{73.5}{1.25} = 58.8 \text{ N}$ *A1*.